

IEEE 830: Software Requirements Specification (SRS)

Callwave Enterprise Video Calling Platform

Document Version: 1.0

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Project: Callwave - Enterprise Video/Audio Calling Platform

Target Audience: Development Team, QA, Product Management, Stakeholders

Status: APPROVED FOR DEVELOPMENT

1. INTRODUCTION

1.1 Purpose

This document specifies the complete functional and non-functional requirements for **Callwave**, an enterprise-grade video and audio calling platform designed for 400,000 users in Cameroon, with initial focus on 20 concurrent users at launch (September 10, 2026).

The SRS defines:

- What the system must do (functional requirements)
- How well it must perform (non-functional requirements)
- Constraints and assumptions
- User stories with acceptance criteria
- Success metrics

This document UNLOCKS all teams - developers, QA, product, and stakeholders will follow these requirements exactly.

1.2 Scope

IN SCOPE (MVP - Must Deliver by Sep 10):

- ✓ Desktop application (Flutter: macOS + Windows)
- ✓ User authentication (desktop signin only)

- ✓ One-on-one video/audio calls
- ✓ Call recording (1-month retention)
- ✓ Contacts management
- ✓ Settings (device, profile, logout)
- ✓ Admin dashboard (user management, analytics, recordings download)
- ✓ Network optimization (Cameroon-specific)

OUT OF SCOPE (Phase 2 - Post-Launch):

- ✗ Web application (comes after desktop is stable)
- ✗ Group meetings (>2 users) - Phase 2
- ✗ Transcription service - Phase 2
- ✗ SIP/PBX integration - Phase 2
- ✗ Calendar integration (built-in only, no external sync) - Phase 2
- ✗ Mobile app - Phase 3

LATER PHASES (Web, Groups, Calendar Sync):

- Phase 2 (Month 3-4): Web application, group calls, external calendar sync
- Phase 3 (Month 5+): Mobile app (iOS/Android), advanced features

1.3 Key Definitions

Term	Definition
Participant	User in an active call
Call	One-on-one or group conversation (video/audio)
Recording	MP4 file of call captured for later playback
Concurrent Users	Number of simultaneous calls happening at same time
Registered Users	Total accounts created on platform
Cameroon Network	Constrained bandwidth (0.5-8 Mbps), high latency (60-100ms+), packet loss (2-5%)
MOS	Mean Opinion Score (ITU-T P.800): 1-5 scale for call quality
RTT	Round-Trip Time: latency between client and server (target: <300ms)

Term	Definition
Admin	Internal team member with elevated privileges
SFU	Selective Forwarding Unit (Mediasoup) - media relay server

1.4 Document Organization

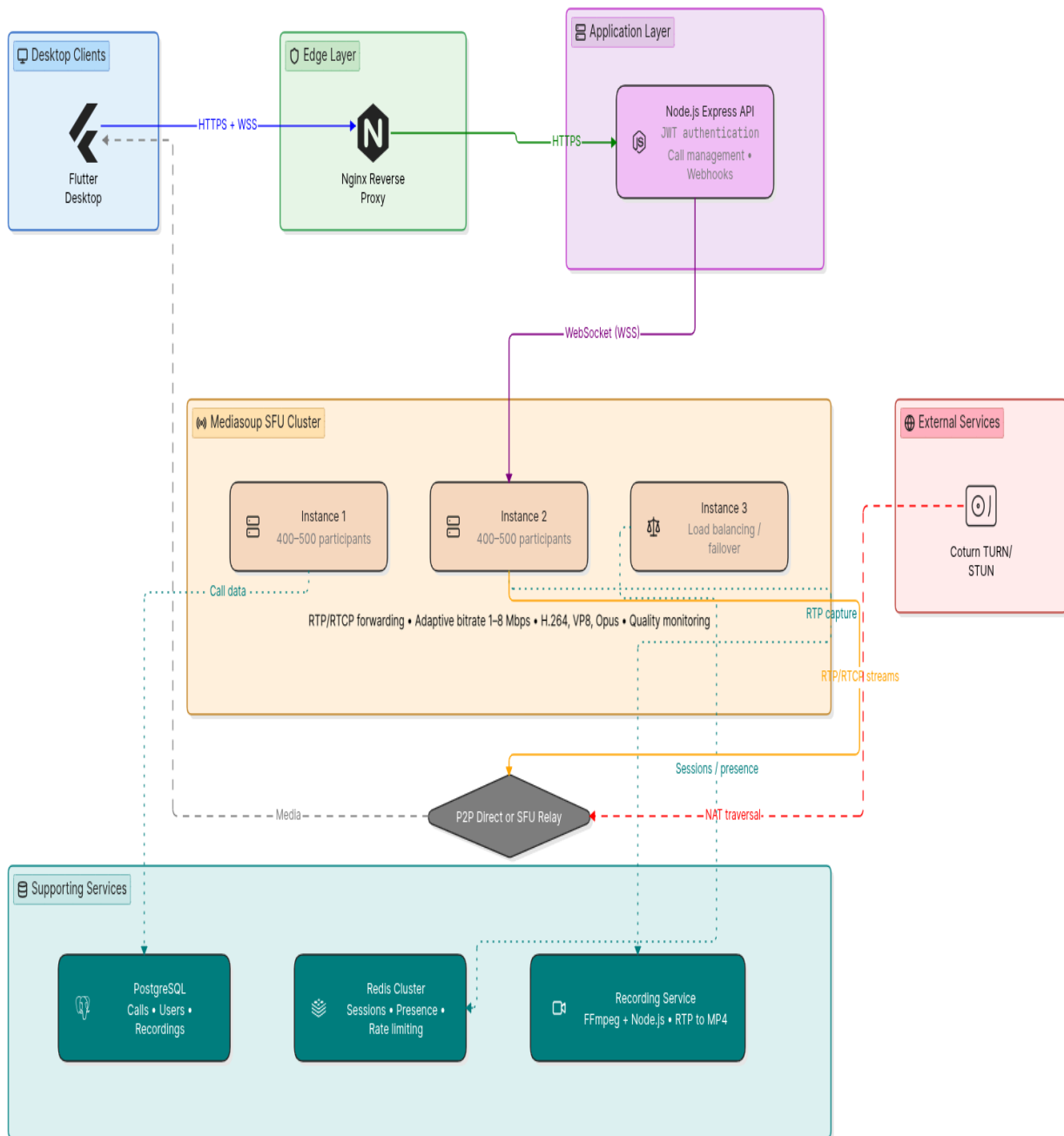
Section	Content
1. Introduction	Purpose, scope, definitions
2. Overall Description	Product perspective, users, constraints
3. Specific Requirements	Functional & non-functional requirements
4. User Stories	Jira-ready user stories with acceptance criteria
5. System Use Cases	Sequence diagrams & workflows
6. Cameroon-Specific Optimizations	Network constraints & solutions
7. Performance Targets	Metrics & success criteria
8. Quality Attributes	Security, availability, usability, scalability

2. OVERALL DESCRIPTION

2.1 Product Perspective

Callwave is a **standalone desktop application** (built with Flutter) that enables secure video/audio calling for Cameroon-based professionals and teams. It operates independently without browser or web dependency at launch.

System Context:



2.2 Product Functions (High-Level)

Function	Description
Authentication	Sign in with email/password (secure)
Call Management	Initiate, receive, end calls
Recording	Automatic or manual call recording to MP4
Contacts	Manage contact list, view call history
Settings	Configure device (camera, mic), profile
Admin Dashboard	Manage users, view analytics, monitor quality
Network Optimization	Automatic bitrate adaptation for poor networks
Notifications	Incoming call alerts, call status updates

2.3 User Classes & Characteristics

User Class 1: Individual Caller

- **Profile:** Cameroon-based professional (employee, entrepreneur)
- **Frequency:** Daily calls (5-10 calls/day)
- **Technical Level:** Basic - expects simple interface
- **Network:** Mobile/WiFi (variable quality)
- **Key Need:** Reliable calling despite poor network

User Class 2: Team Admin

- **Profile:** Internal Callwave team managing platform
- **Frequency:** Daily (monitor calls, user management)
- **Technical Level:** Intermediate - understands networking
- **Key Need:** Visibility into platform health, user management

User Class 3: System Administrator

- **Profile:** Customer enterprise admins (Phase 2+)
- **Frequency:** Daily
- **Technical Level:** Advanced - manage users, billing, compliance
- **Key Need:** User management, analytics, security controls

2.4 Operating Environment

Desktop Environment (MVP):

- **OS:** macOS 10.15+ (Intel & Apple Silicon), Windows 10/11, Linux(debian)
- **Network:** WiFi or mobile hotspot (often poor quality)
- **Flutter Version:** Latest stable (3.22+)
- **Connectivity:** HTTPS + WebSocket (WSS)

Backend Infrastructure:

- **Server OS:** Linux (Ubuntu 20.04 LTS or later)
- **Node.js:** v18+
- **Database:** PostgreSQL 13+
- **Cache:** Redis 6+
- **Media Server:** Mediasoup (latest)
- **NAT Traversal:** Coturn (self-hosted in Cameroon)

Network Conditions (Cameroon-Specific):

- **Bandwidth:** 0.5 - 8 Mbps (average: 2.5 Mbps)
- **Latency:** 60-100ms+ (due to geographic distance)
- **Packet Loss:** 2-5% (common)
- **Jitter:** 20-50ms (variable delay)

2.5 System Constraints

Technical Constraints:

1. **No external authentication** (OAuth/Google) for MVP - desktop only
2. **Recordings stored on server** (not cloud) - 1-month retention
3. **Single datacenter** (Cameroon-based) - no geographic failover for now
4. **Desktop-only UI** (web/mobile = Phase 2+)
5. **Open-source media stack** (Mediasoup, not proprietary)

Business Constraints:

1. **Timeline:** 2-month sprint cycle (4 sprints = 8 weeks)
2. **Launch date:** September 10, 2026 (hard deadline)
3. **Initial users:** 20 concurrent at launch (scale gradually)
4. **Team size:** Small dev team (efficient, focused)
5. **Budget:** No unlimited resources (optimize infrastructure costs)

Regulatory/Compliance:

1. **Encryption:** All media encrypted end-to-end (SRTP per RFC 3711)
 2. **Data retention:** Call logs & recordings retained per compliance requirements
 3. **User privacy:** No data sharing with third parties
 4. **GDPR-adjacent:** Support user data export/deletion (Phase 2)
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3. SPECIFIC REQUIREMENTS

3.1 FUNCTIONAL REQUIREMENTS

3.1.1 Authentication & User Management

REQ-AUTH-001: User Signup

- **Priority:** CRITICAL
- **Description:** New user can create account with email, matricule & password
- **Actors:** Unregistered user
- **Preconditions:** User has valid matricule
- **Steps:**
 1. User launches Callwave desktop app
 2. Clicks "Sign Up" button
 3. Enters matricule, email
 4. Enters password (min 8 chars, alphanumeric + special char)
 5. Confirms password
 6. Clicks "Create Account"
 7. Server creates user in PostgreSQL
 8. System sends verification email (Phase 2 - for now auto-verified)
- **Postconditions:** Account created, user logged in, redirected to home screen
- **Acceptance Criteria:**
 - ☒ Email format validated (RFC 5322)
 - ☒ Password strength enforced (min 8 chars, 1 uppercase, 1 number, 1 special)
 - ☒ Duplicate email rejected
 - ☒ Duplicate matricule rejected
 - ☒ Response time: <2 seconds
 - ☒ User session created automatically
 - ☒ Secure password hashing (bcrypt, not plaintext)

REQ-AUTH-002: User Login

- **Priority:** CRITICAL
- **Description:** Existing user logs in with email & password
- **Actors:** Registered user
- **Preconditions:** User account exists
- **Steps:**
 1. User opens Callwave app
 2. Enters email and password
 3. Clicks "Login"
 4. Server validates credentials against database
 5. If valid: create session token (JWT)
 6. Return token to client
 7. Client stores token securely (encrypted)
 8. Redirect to home screen
- **Postconditions:** User logged in, can make calls
- **Acceptance Criteria:**
 - ☒ Invalid credentials show error (don't reveal if email exists)
 - ☒ Session token expires after 24 hours
 - ☒ Token stored securely (no plaintext in logs)
 - ☒ Response time: <2 seconds
 - ☒ Failed login attempt logged (for security audit)
 - ☒ Rate limiting: Max 5 failed attempts per email per 15 minutes

REQ-AUTH-003: User Logout

- **Priority:** HIGH
- **Description:** User logs out, session ends
- **Postconditions:** Session invalidated, redirected to login screen
- **Acceptance Criteria:**
 - ☒ Token revoked on server
 - ☒ Client clears cached token
 - ☒ User cannot make calls after logout
 - ☒ Logout completes in <1 second

REQ-AUTH-004: Profile Management

- **Priority:** MEDIUM
- **Description:** User can view/edit profile (name, photo, status)
- **Acceptance Criteria:**
 - ☒ User can update display name
 - ☒ User can upload profile photo (max 2MB, JPEG/PNG)
 - ☒ User can change password (requires old password verification)
 - ☒ Changes saved to database within <2 seconds

3.1.2 Call Management (One-on-One)

REQ-CALL-001: Initiate Call

- **Priority:** CRITICAL
- **Description:** User can start a call with another user
- **Actors:** Caller
- **Preconditions:** Caller logged in, recipient exists in contacts
- **Steps:**
 1. User selects contact from list (or searches by email/name)
 2. Clicks "Call" button (video or audio icon)
 3. Client sends call invitation via WebSocket to backend
 4. Backend routes invitation to recipient (if online)
 5. Recipient receives incoming call notification
 6. Recipient accepts/declines call
 7. If accepted: establish WebRTC connection to SFU
 8. Media streams begin flowing through Mediasoup
- **Postconditions:** Call active, both participants can see/hear
- **Acceptance Criteria:**
 - ☒ Invitation sent within <1 second
 - ☒ Recipient notified within <2 seconds (if online)
 - ☒ Call establishment time: <5 seconds (connection setup)
 - ☒ Offline recipient: Call invitation stored for retry when online
 - ☒ Outgoing call shows calling UI (cancel button, timeout after 30 seconds)
 - ☒ Both participants recorded automatically (can disable in settings)

REQ-CALL-002: Receive Call

- **Priority:** CRITICAL
- **Description:** User receives incoming call notification and can accept/decline
- **Actors:** Recipient
- **Preconditions:** Recipient logged in
- **Steps:**
 1. Incoming call notification displayed (sound + visual alert)
 2. User sees caller name/photo
 3. User clicks "Accept" (video/audio) or "Decline"
 4. If accept: establish connection to SFU
 5. If decline: call ended, caller notified
- **Postconditions:** Call active (if accepted) or ended (if declined)
- **Acceptance Criteria:**
 - ☒ Notification appears within <2 seconds of invitation

- ☒ Ringtone plays (can mute in settings)
- ☒ Notification persists until answered or timeout (30 seconds)
- ☒ Declining call is instant (<500ms)
- ☒ Missed call logged in call history

REQ-CALL-003: End Call

- **Priority:** CRITICAL
- **Description:** User can end call
- **Postconditions:** Call terminated, media streams closed, recording finalized
- **Acceptance Criteria:**
 - ☒ End call button available during call
 - ☒ Clicking end call terminates connection within <1 second
 - ☒ Both participants notified call ended
 - ☒ Recording stopped and saved
 - ☒ Call duration calculated and logged
 - ☒ Resources released (camera, mic released for other apps)

REQ-CALL-004: Mute/Unmute Microphone

- **Priority:** HIGH
- **Description:** User can mute/unmute audio during call
- **Acceptance Criteria:**
 - ☒ Mute button visible during call
 - ☒ Muting takes effect within <500ms
 - ☒ Other participant notified of mute status
 - ☒ Muted indicator shown on screen
 - ☒ Audio not sent to recipient when muted

REQ-CALL-005: Enable/Disable Video

- **Priority:** HIGH
- **Description:** User can turn video on/off during call
- **Acceptance Criteria:**
 - ☒ Video toggle button visible during call
 - ☒ Video disable takes effect within <1 second
 - ☒ Other participant sees "Video Off" placeholder
 - ☒ Bandwidth reduced when video disabled (important for poor networks)
 - ☒ Can re-enable video at any time

REQ-CALL-006: Call Duration Timer

- **Priority:** MEDIUM
- **Description:** Show elapsed time during call
- **Acceptance Criteria:**
 - ☒ Timer displayed in HH:MM:SS format

- ☒ Timer updates every second
 - ☒ Accurate within ± 1 second
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3.1.3 Recording

REQ-REC-001: Automatic Call Recording

- **Priority:** HIGH
- **Description:** Calls are automatically recorded to MP4 file
- **Acceptance Criteria:**
 - ☒ Recording starts automatically when call connects
 - ☒ Recording stops when call ends
 - ☒ Format: MP4 with H.264 video + Opus audio
 - ☒ Codec: H.264 Baseline profile, 90kHz (video); Opus 48kHz (audio)
 - ☒ Bitrate: 3000 kbps (optimized for playback quality)
 - ☒ Resolution: 720p (1280x720) on good network, adaptive lower on poor network
 - ☒ File saved to server with timestamp:
`recording_USERID_CALLID_YYYYMMDD_HHMMSS.mp4`
 - ☒ Recording linked to both participants' accounts
 - ☒ Storage location: `/data/recordings/` on server (managed by ops)

REQ-REC-002: Recording Retention Policy

- **Priority:** HIGH
- **Description:** Recordings automatically deleted after 1 month
- **Acceptance Criteria:**
 - ☒ Recordings retained for exactly 30 calendar days
 - ☒ Expiration date calculated at creation time
 - ☒ Automatic cleanup job runs daily (via cron) to delete expired recordings
 - ☒ Deletion logged in audit trail
 - ☒ User notified when recording approaching expiration (7 days before)
 - ☒ Users can manually delete recordings before expiration

REQ-REC-003: Recording Disable Option

- **Priority:** MEDIUM
- **Description:** User can disable recording in settings (per-device)
- **Acceptance Criteria:**
 - ☒ Setting in Settings → Recording
 - ☒ Toggle: "Record my calls"
 - ☒ Default: ON (record by default)

- ☒ If disabled: no MP4 file created
 - ☒ Other participant still notified (transparency)
 - ☒ Setting persists across app restarts
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3.1.4 Contacts Management

REQ-CONT-001: View Contacts

- **Priority:** HIGH
- **Description:** User can view list of contacts
- **Acceptance Criteria:**
 - ☒ Contacts list displayed in UI
 - ☒ Shows: Name, email, last call date, status (online/offline)
 - ☒ Sorted alphabetically by first name
 - ☒ Search box to find contacts by name/email
 - ☒ Search results within <500ms

REQ-CONT-002: Add Contact

- **Priority:** HIGH
- **Description:** User can add contact manually
- **Acceptance Criteria:**
 - ☒ "Add Contact" button opens dialog
 - ☒ Enter email address of person to add
 - ☒ Validation: Email must exist in system
 - ☒ Contact added to list
 - ☒ Other person notified of follow (can allow/deny in Phase 2)

REQ-CONT-003: Call History

- **Priority:** HIGH
 - **Description:** User can view call history (incoming, outgoing, missed)
 - **Acceptance Criteria:**
 - ☒ Call history shows: Date, time, duration, participant name, type (incoming/outgoing/missed)
 - ☒ Sorted by date (newest first)
 - ☒ Missed calls indicated with badge
 - ☒ Click on call history item to see details
 - ☒ History retained for 6 months (compliance)
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3.1.5 Settings

REQ-SET-001: Device Settings

- **Priority:** HIGH
- **Description:** User can configure camera and microphone
- **Acceptance Criteria:**
 - ☒ Settings → Devices
 - ☒ List available cameras (with preview)
 - ☒ List available microphones (with audio level indicator)
 - ☒ List available speakers/output devices
 - ☒ Test audio: Record 5 seconds, play back to test
 - ☒ Changes take effect immediately
 - ☒ Settings persisted to local storage

REQ-SET-002: Audio/Video Quality Settings

- **Priority:** MEDIUM
- **Description:** User can adjust quality preferences
- **Acceptance Criteria:**
 - ☒ Settings → Call Quality
 - ☒ Options: Low (240p), Medium (480p), High (720p), Adaptive (auto)
 - ☒ Default: Adaptive (recommended)
 - ☒ Changes take effect on next call
 - ☒ Persistent across sessions

REQ-SET-003: Notifications Settings

- **Priority:** MEDIUM
- **Description:** User can configure call notifications
- **Acceptance Criteria:**
 - ☒ Toggle: Incoming call alerts (sound/visual)
 - ☒ Toggle: Call status updates
 - ☒ Volume control for ringtone
 - ☒ Default: All enabled

REQ-SET-004: Account Settings

- **Priority:** HIGH
- **Description:** User can manage account
- **Acceptance Criteria:**
 - ☒ View account info: Email, registration date, storage used
 - ☒ Change password
 - ☒ Logout all devices

- ☒ Export call history (CSV format)
 - ☒ Delete account (Phase 2 - GDPR compliance)
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3.1.6 Admin Dashboard

REQ-ADMIN-001: User Management

- **Priority:** CRITICAL
- **Description:** Admin can view, create, suspend, delete users
- **Actors:** Internal admin (Callwave team)
- **Acceptance Criteria:**
 - ☒ Admin panel accessible via Settings (admin-only access)
 - ☒ User list: Email, name, status (active/suspended), registration date, last login
 - ☒ Search users by email/name
 - ☒ Create new user (email, temporary password)
 - ☒ Suspend user (prevent calls, but retain data)
 - ☒ Reactivate suspended user
 - ☒ Delete user (remove from system, keep recordings for legal)
 - ☒ Bulk actions: Suspend/delete multiple users
 - ☒ Audit log: Track all admin actions

REQ-ADMIN-002: Call History & Metrics

- **Priority:** CRITICAL
- **Description:** Admin can view all calls and metrics
- **Acceptance Criteria:**
 - ☒ View all calls: Date, time, caller, callee, duration, status (completed/missed/declined)
 - ☒ Filter by date range, user, status
 - ☒ Export call history to CSV
 - ☒ Metrics dashboard:
 - Total calls (all-time, today, this week)
 - Average call duration
 - Peak concurrent users (timestamp)
 - Success rate (calls that completed / total initiated)
 - ☒ Real-time stats: Current active calls, participants

REQ-ADMIN-003: Recording Management

- **Priority:** HIGH
- **Description:** Admin can download and delete recordings
- **Acceptance Criteria:**

- ☒ View list of recordings: Date, participants, duration, file size
- ☒ Search recordings by participant name, date range
- ☒ Download recording to local disk (MP4)
- ☒ Delete recording manually (before 30-day expiration)
- ☒ Bulk download (zip multiple recordings)

REQ-ADMIN-004: Network Analytics

- **Priority:** HIGH
- **Description:** Admin can view network quality metrics
- **Acceptance Criteria:**
 - ☒ Dashboard showing per-call metrics:
 - RTT (round-trip time): Target <300ms
 - Jitter: Target <30ms
 - Packet loss: % (target <1%)
 - MOS (Mean Opinion Score): ITU-T P.800, target >4.0
 - ☒ Graphs: Hourly/daily trends
 - ☒ Alerts: Flag calls with poor quality (loss >3%, jitter >50ms, RTT >300ms)
 - ☒ Per-user analytics: See each participant's network conditions
 - ☒ Export: CSV with all metrics

REQ-ADMIN-005: Bandwidth Management

- **Priority:** MEDIUM
- **Description:** Admin can set bandwidth limits per user
- **Acceptance Criteria:**
 - ☒ User settings → Bandwidth limit
 - ☒ Options: Unlimited, 8 Mbps, 4 Mbps, 2 Mbps, 1 Mbps (for poor networks)
 - ☒ Default: Unlimited (client-side adaptive algorithm still limits)
 - ☒ Limit enforced by SFU (producer max bitrate)
 - ☒ User notified if they exceed limit during call

REQ-ADMIN-006: Billing Data

- **Priority:** MEDIUM
 - **Description:** Admin can view/export billing data (Phase 1 basic, Phase 2 advanced)
 - **Acceptance Criteria:**
 - ☒ Total users: Registered, active, suspended
 - ☒ Call volume: Total calls, total minutes, peak concurrent
 - ☒ Recording storage: GB used, storage limit
 - ☒ Export: CSV format for accounting/billing system
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3.2 NON-FUNCTIONAL REQUIREMENTS

3.2.1 Performance

REQ-PERF-001: Call Establishment Time

- **Requirement:** Call connection establishes within 5 seconds
- **Measurement:** Time from "Call" button click to first media frame displayed
- **Acceptance Criteria:**
 -  RTT to backend: <2 seconds
 -  ICE candidate gathering: <3 seconds
 -  WebRTC connection: <2 seconds
 -  Total: <5 seconds (95th percentile)
 -  Test on: Good network (10 Mbps), poor network (1 Mbps)

REQ-PERF-002: Latency Target

- **Requirement:** Round-trip time <300ms (one-way <150ms)
- **Measurement:** RTT measured via RTCP reports
- **Cameroon Context:** Due to geographic distance, 60-100ms baseline acceptable, <150ms one-way target
- **Acceptance Criteria:**
 -  95th percentile: <300ms RTT
 -  99th percentile: <400ms RTT (acceptable but degraded)
 -  Monitor via admin dashboard
 -  Alert if >300ms sustained for >5 minutes

REQ-PERF-003: Jitter Target

- **Requirement:** Jitter <30ms for smooth audio/video
- **Measurement:** Variance in inter-packet delay (via RTCP)
- **Acceptance Criteria:**
 -  95th percentile: <30ms
 -  99th percentile: <50ms
 -  Detected via RTP timestamp analysis
 -  High jitter triggers quality degradation warnings

REQ-PERF-004: Adaptive Bitrate (Cameroon Critical)

- **Requirement:** System automatically adapts video bitrate based on network
- **Measurement:** Video bitrate adjustment every 2-5 seconds based on congestion feedback
- **Acceptance Criteria:**
 -  Bitrate targets (RFC 8083 congestion control):

- <1 Mbps available: 800 kbps video
- 1-2 Mbps: 1.5 Mbps
- 2-5 Mbps: 3 Mbps
 - 5 Mbps: 5 Mbps (max for now)
- ☒ Quality reduction smooth (no jarring jumps)
- ☒ Resolution reduction: 1280x720 → 854x480 → 426x240
- ☒ Framerate reduction: 30fps → 15fps (if needed)
- ☒ Algorithm: Transport-wide CC (RFC 8083)
- ☒ Bitrate measured via WebRTC stats API

REQ-PERF-005: Packet Loss Resilience

- **Requirement:** Call quality maintained with 2-5% packet loss (Cameroon-typical)
- **Acceptance Criteria:**
 - ☒ FEC (Forward Error Correction): Enable redundancy (RFC 3711 SRTP)
 - ☒ Retransmission: Audio retransmits lost packets automatically
 - ☒ Video: Handles loss gracefully (show previous frame if needed)
 - ☒ Test: Simulate 3% loss → call still usable
 - ☒ Test: Simulate 5% loss → quality reduced but not dropped

REQ-PERF-006: Response Time

- **Requirement:** UI interactions responsive (<500ms latency perceived)
- **Acceptance Criteria:**
 - ☒ Button clicks: Feedback within <100ms
 - ☒ Search results: <500ms to display
 - ☒ Admin dashboard load: <2 seconds
 - ☒ Settings save: <1 second

3.2.2 Security

REQ-SEC-001: End-to-End Encryption (CRITICAL)

- **Requirement:** All media encrypted SRTP (Secure RTP)
- **Specification:** RFC 3711, RFC 5346 (DTLS-SRTP key agreement)
- **Cipher Suite:** AES-128-CM (Counter Mode) + HMAC-SHA1
- **Acceptance Criteria:**
 - ☒ Media packets encrypted before transmission
 - ☒ Keys exchanged via DTLS handshake (TLS for UDP)
 - ☒ Encryption verified via packet inspection (no plaintext RTP)
 - ☒ Forward Secrecy: Compromise of one call doesn't affect others
 - ☒ Test: Wireshark packet capture shows encrypted payload

REQ-SEC-002: Transport Encryption

- **Requirement:** All control signaling encrypted TLS 1.3+
- **Specification:** HTTPS for REST, WSS (WebSocket Secure) for signaling
- **Acceptance Criteria:**
 - ☒ All HTTP → HTTPS (no plaintext)
 - ☒ All WebSocket → WSS (encrypted)
 - ☒ TLS 1.3 minimum (no legacy TLS 1.0-1.2)
 - ☒ Certificate: Valid, signed by trusted CA
 - ☒ HSTS header enforced (require HTTPS)

REQ-SEC-003: Password Security

- **Requirement:** Passwords hashed securely, never stored plaintext
- **Specification:** bcrypt with salt, min cost factor 12
- **Acceptance Criteria:**
 - ☒ Passwords hashed with bcrypt (not MD5, SHA1)
 - ☒ Salt: Random per user
 - ☒ Cost factor: ≥ 12 (slow to brute-force)
 - ☒ Never log or display passwords
 - ☒ Password reset: Send token link (not password via email)

REQ-SEC-004: Authentication Token Security

- **Requirement:** Session tokens secure, short-lived, XSS-protected
- **Specification:** JWT with HMAC-SHA256 signing
- **Acceptance Criteria:**
 - ☒ Token format: JWT (JSON Web Token)
 - ☒ Expiration: 24 hours (force re-login daily)
 - ☒ Storage: Secure storage (not localStorage if possible, encrypted)
 - ☒ Refresh token: Implement refresh mechanism for long sessions (Phase 2)
 - ☒ Revocation: Token invalidated on logout

REQ-SEC-005: Rate Limiting

- **Requirement:** Prevent brute-force attacks
- **Acceptance Criteria:**
 - ☒ Login attempts: Max 5 failures per email per 15 minutes
 - ☒ API calls: Max 100 requests per minute per user
 - ☒ Response: HTTP 429 (Too Many Requests) with retry-after header
 - ☒ Failed attempt logging: Log for security audit

REQ-SEC-006: Input Validation & SQL Injection Prevention

- **Requirement:** All user input validated, SQL injection impossible

- **Acceptance Criteria:**
 - ☒ Email: Validated against RFC 5322
 - ☒ Names: Max 100 chars, no special chars except hyphen/apostrophe
 - ☒ Passwords: Validated on client and server
 - ☒ Database queries: Use parameterized statements (no string concatenation)
 - ☒ API inputs: Schema validation (e.g., JSON Schema)

REQ-SEC-007: Audit Logging

- **Requirement:** All security events logged for investigation
- **Acceptance Criteria:**
 - ☒ Logins: Success and failure
 - ☒ Failed attempts: Timestamp, IP, email
 - ☒ Admin actions: User creation/deletion, suspension
 - ☒ Password changes: Timestamp
 - ☒ Retention: Logs kept for 6 months
 - ☒ Access control: Only admins can view audit logs

REQ-SEC-008: Data Privacy

- **Requirement:** User data not shared with third parties
 - **Acceptance Criteria:**
 - ☒ No analytics tracking (no Google Analytics, Mixpanel, etc.)
 - ☒ No ads or third-party cookies
 - ☒ Call recordings stored securely on-premises
 - ☒ User data export available (JSON format) - Phase 2
 - ☒ Account deletion removes all user data - Phase 2
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3.2.3 Availability & Reliability

REQ-AVAIL-001: System Uptime

- **Requirement:** Platform available 99.5% of the time
- **Measurement:** $(\text{Total minutes in period} - \text{Downtime}) / \text{Total minutes} * 100$
- **Target:** 99.5% (43 minutes downtime/month acceptable)
- **Acceptance Criteria:**
 - ☒ Monitoring: Uptime tracking via external heartbeat checks
 - ☒ Alerts: Immediate notification if service down >5 minutes
 - ☒ SLA: 99.5% commitment published to users
 - ☒ Status page: <https://status.callwave.cm> shows real-time status
 - ☒ Maintenance windows: Scheduled only, communicated 7 days in advance

REQ-AVAIL-002: Graceful Degradation

- **Requirement:** If call drops, system recovers automatically or notifies user
- **Acceptance Criteria:**
 - ☒ Network connection loss detected within <2 seconds
 - ☒ Auto-reconnection attempted (exponential backoff: 1s, 2s, 4s, 8s, max 30s)
 - ☒ User notified: "Reconnecting..." toast message
 - ☒ If reconnection fails: "Call ended - poor connection" message
 - ☒ Automatic call recording save on disconnect

REQ-AVAIL-003: Database Backup & Recovery

- **Requirement:** User data backed up and recoverable
- **Acceptance Criteria:**
 - ☒ Daily database backups (automated)
 - ☒ Backup retention: 30 days
 - ☒ Recovery time objective (RTO): 1 hour (restore within 1 hour)
 - ☒ Recovery point objective (RPO): <1 hour data loss acceptable
 - ☒ Backup stored separately from primary database (different server)

REQ-AVAIL-004: Error Handling

- **Requirement:** Errors handled gracefully, don't crash app
 - **Acceptance Criteria:**
 - ☒ Try-catch blocks around all network calls
 - ☒ User sees friendly error messages (not stacktraces)
 - ☒ Errors logged to server for debugging
 - ☒ Retry logic: Auto-retry failed requests 3 times before showing error
-

3.2.4 Scalability

REQ-SCAL-001: Concurrent Users

- **Requirement:** System supports 20 concurrent users at launch, scale to 1000+ eventually
- **Acceptance Criteria:**
 - ☒ Launch (Sep 10): Support 20 concurrent calls
 - ☒ Phase 2: Scale to 100 concurrent calls
 - ☒ Year 1: Support 1,000 concurrent calls
 - ☒ Architecture supports horizontal scaling (add more SFU instances)
 - ☒ Load testing: Verify performance at 20, 100, 1000 concurrent users

-  Degradation acceptable: Quality may reduce at peak load, but calls stay connected

REQ-SCAL-002: Database Scaling

- **Requirement:** Database scales with user growth (from 20 to 400k users)
- **Acceptance Criteria:**
 -  PostgreSQL: Indexing on frequently queried columns (email, user_id, call_id)
 -  Query optimization: < 500ms response time for admin queries
 -  Sharding: Plan for horizontal partition if >100k users (Phase 2)
 -  Caching: Redis for session/presence data (fast lookups)

REQ-SCAL-003: Storage Scaling

- **Requirement:** Storage for recordings grows with call volume
 - **Acceptance Criteria:**
 -  Estimated growth: ~500MB per 1000 calls (MP4, 1-min avg call)
 -  Retention: 1 month (auto-delete older files)
 -  Expansion: Plan storage expansion before hitting 80% capacity
 -  Monitoring: Alert when storage >80% full
-

3.2.5 Usability

REQ-USAB-001: Simple Interface

- **Requirement:** UI intuitive, no training required
- **Acceptance Criteria:**
 -  Home screen: 3 main buttons (Contacts, Call History, Settings)
 -  Call screen: 4 controls (Mute, Video, End, More)
 -  Consistent icons: Reuse standard icons (phone, video, settings, etc.)
 -  Dark/Light mode: Support both (user preference)
 -  Accessibility: Support screen readers (Phase 2)

REQ-USAB-002: Responsiveness

- **Requirement:** UI responds instantly to user actions
- **Acceptance Criteria:**
 -  Button clicks show visual feedback within <100ms
 -  Transitions smooth (not jarring)
 -  No frozen UI during network operations
 -  Loading indicators shown for long operations (>2 seconds)

REQ-USAB-003: Internationalization (Phase 2)

- **Requirement:** Support English + French (Cameroon official languages)
- **Acceptance Criteria:**
 - ☒ English as default
 - ☒ French translations (Phase 2)
 - ☒ Language setting in Settings
 - ☒ All UI text translatable (no hardcoded strings)

REQ-USAB-004: Documentation

- **Requirement:** Users can get help
 - **Acceptance Criteria:**
 - ☒ In-app Help button → FAQ
 - ☒ FAQ covers: How to call, record, manage contacts, troubleshoot
 - ☒ Email support: support@callwave.cm
 - ☒ Video tutorials (Phase 2)
-

3.3 INTERFACE REQUIREMENTS

3.3.1 User Interface (Desktop Flutter)

Main Screens:

1. Login Screen

- Email input field
- Password input field
- "Login" button
- "Sign Up" link
- Remember me checkbox (Phase 2)

2. Home Screen (After Login)

- Tabs: Contacts | Call History | Settings
- Quick call button (large, prominent)
- User status (online/offline badge)
- Notifications icon (missed calls count)

3. Contacts Screen

- Contact list
- Search box
- Add contact button (+)
- Each contact: Name, email, last call date, online status

4. **Call History Screen**

- List of calls: Date, time, caller/callee, duration, status badge
- Filter: All | Missed | Incoming | Outgoing
- Search box

5. **Active Call Screen**

- Remote participant video (large)
- Local video preview (small corner)
- Mute button (toggles mic)
- Video button (toggles camera)
- End call button (red, prominent)
- Call timer (HH:MM:SS)
- More options menu (screen share - Phase 2)

6. **Settings Screen**

- Devices (camera, mic, speakers)
- Call quality (Low/Medium/High/Adaptive)
- Notifications (toggles)
- About (version, legal)
- Logout button

7. **Admin Dashboard** (Admin-only, accessible via Settings)

- Users tab: User list, create/suspend/delete
- Calls tab: Call history, metrics, export
- Recordings tab: Recording list, download, delete
- Analytics tab: Network metrics (latency, jitter, MOS, loss)
- Bandwidth tab: User limits

3.3.2 API Interfaces (Backend)

REST Endpoints:

Authentication:

POST /api/v1/auth/signup → Create account

POST /api/v1/auth/login → Login, return JWT token

POST /api/v1/auth/logout → Logout, revoke token

POST /api/v1/auth/refresh → Refresh token (Phase 2)

User Profile:

GET /api/v1/users/me → Get logged-in user info

PUT /api/v1/users/me → Update profile

POST /api/v1/users/me/password → Change password

Contacts:

GET /api/v1/contacts → List contacts

POST /api/v1/contacts → Add contact

GET /api/v1/contacts/{id} → Get contact details

DELETE /api/v1/contacts/{id} → Remove contact

Calls:

POST /api/v1/calls → Initiate call

GET /api/v1/calls → Get call history

GET /api/v1/calls/{id} → Get call details

POST /api/v1/calls/{id}/end → End call

Recordings:

GET /api/v1/recordings → List recordings

GET /api/v1/recordings/{id} → Download recording (streaming)

DELETE /api/v1/recordings/{id} → Delete recording

Admin:

GET /api/v1/admin/users → List all users (admin only)

POST /api/v1/admin/users → Create user (admin only)

PUT /api/v1/admin/users/{id} → Suspend/reactivate (admin only)

DELETE /api/v1/admin/users/{id} → Delete user (admin only)

GET /api/v1/admin/metrics → Get platform metrics (admin only)

GET /api/v1/admin/analytics → Get network analytics (admin only)

WebSocket Events (Real-time Signaling):

Incoming:

call:invite → Incoming call invitation {from, callId}

call:accept → Peer accepted call

call:decline → Peer declined call

call:end → Peer ended call

message:* → Chat messages (Phase 2)

Outgoing:

call:initiate → Send call invitation

call:accept → Accept incoming call

call:decline → Decline call

call:end → End call

user:typing → Typing indicator (Phase 2)

Mediasoup WebRTC Signaling:

Client → Server:

getRouterRtpCapabilities → Request router capabilities

createWebRtcTransport → Create send/recv transports

produce → Create producer (video/audio)

consume → Create consumer (receive video/audio)

connectWebRtcTransport → Exchange DTLS parameters

Server → Client:

rtpCapabilities → Router capabilities

webRtcTransportOptions → Transport params (IP, port, DTLSParams)

producerAdded → Peer produced media

consumerCreated → New consumer ready for stats

3.4 CAMEROON-SPECIFIC REQUIREMENTS

3.4.1 Network Constraints

REQ-CM-001: Bandwidth Optimization

- **Constraint:** Cameroon average bandwidth 2.5 Mbps (range: 0.5-8 Mbps)
- **Impact:** Cannot support full HD (1080p @ 30fps needs 5+ Mbps)
- **Solution:**
 -  Adaptive bitrate: Reduce resolution/framerate on low bandwidth
 -  Default quality: 720p @ 30fps (requires 3 Mbps)
 -  Poor network fallback: 480p @ 15fps (needs 1.5 Mbps)
 -  Emergency fallback: Audio-only (requires 0.1 Mbps)

REQ-CM-002: Latency Handling

- **Constraint:** Cameroon latency 60-100ms+ (geographic distance to servers)
- **Impact:** Users perceive 120-200ms delay (2-way), which is noticeable but acceptable
- **Solution:**
 -  Server location: Host in Cameroon (or nearest African datacenter)

-  Anticipation UI: Show "connecting..." to manage expectations
-  Target: Keep one-way latency <150ms (aggressive optimization)

REQ-CM-003: Packet Loss Resilience (2-5% Common)

- **Constraint:** Cameroon networks lose 2-5% of packets (unstable ISP routes)
- **Impact:** Video/audio quality degradation without mitigation
- **Solution:**
 -  FEC (Forward Error Correction): Add redundancy to packets
 -  Audio redundancy: RFC 2198 (retransmit audio frames)
 -  Video: Use intra-frame refresh (send I-frames when loss detected)
 -  Test: Verify call quality with 3-5% simulated loss

REQ-CM-004: Jitter Handling

- **Constraint:** Variable network delays cause jitter (delay variance)
- **Impact:** Audio clicks/pops, video freezes
- **Solution:**
 -  Jitter buffer: Store incoming packets, release at constant rate
 -  Target jitter <30ms (95th percentile)
 -  Adaptive buffer: Increase depth if jitter >50ms (adds latency but improves audio)

3.4.2 Data Compression

REQ-CM-005: Efficient Encoding

- **Requirement:** Minimize bandwidth usage
- **Acceptance Criteria:**
 -  H.264 Baseline Profile: More compression (better for poor networks)
 -  Opus audio: 6-128 kbps adaptive (vs G.711 64 kbps fixed)
 -  Video resolution reduction: Auto-drop from 720p to 480p to 240p
 -  Framerate reduction: 30fps → 15fps on poor networks
 -  No WebP/HEVC (H.265): Lower device support in Cameroon

3.4.3 Offline Message Queue

REQ-CM-006: Offline Message Handling

- **Requirement:** Handle temporary disconnects gracefully
- **Acceptance Criteria:**
 -  If network drops during call: Auto-reconnect within 30 seconds
 -  If recipient offline when call initiated: Queue invitation, retry when online
 -  Retry queue: Try 5 times over 5 minutes before giving up
 -  Offline notifications: Deliver when user comes online

- ☒ User notified: "Trying to reach recipient..." message

3.4.4 Low-End Device Support

REQ-CM-007: Minimum Device Requirements

- **Requirement:** Support older/mid-range computers common in Cameroon
- **Acceptance Criteria:**
 - ☒ Minimum RAM: 2GB (can call with 4GB Flutter + browser overhead)
 - ☒ Minimum CPU: Dual-core (H.264 decoding requires some power)
 - ☒ Storage: 100MB for app + 500MB for recordings cache
 - ☒ Battery: Optimize to not drain battery quickly (disable auto-recording if option available - Phase 2)

3.5 CAMEROON NETWORK CONDITIONS - TEST SCENARIOS

All of these scenarios must be tested before launch:

Scenario	Bandwidth	RTT	Jitter	Loss	Outcome
Excellent	5+ Mbps	<50ms	<10ms	<0.1%	Full HD video, smooth audio
Good	2-5 Mbps	80ms	<20ms	<1%	720p video, good audio
Fair	1-2 Mbps	100ms	30ms	2%	480p video, acceptable audio
Poor	0.5-1 Mbps	150ms	50ms	5%	240p video, audio may be choppy
Very Poor	<0.5 Mbps	200ms+	100ms	5%+	Audio only, or call fails

4. USER STORIES (JIRA-Ready Format)

User Story Template

User Story ID: US-XXX

Title: [Brief description]

Priority: CRITICAL / HIGH / MEDIUM / LOW

Story Points: [1, 2, 3, 5, 8, 13]

Acceptance Criteria: [Given/When/Then format]

Definition of Done: Coded + Unit tested + Code reviewed + Merged to main + Documented

User Stories (Selected Key Ones)

US-001: User Registration

Title: As an unregistered user, I want to create an account so that I can use Callwave

Priority: CRITICAL

Story Points: 5

Acceptance Criteria:

- Given: User opens Callwave app
- When: Clicks "Sign Up"
- Then: Form displayed with email, password, confirm password fields
- Given: User enters valid email and password (8+ chars, 1 uppercase, 1 number, 1 special)
- When: Clicks "Create Account"
- Then: Account created, user logged in, home screen displayed
- Given: User enters duplicate email
- When: Clicks "Create Account"

- Then: Error shown "Email already registered"
- Given: User enters weak password (less than 8 chars)
- When: Clicks "Create Account"
- Then: Error shown "Password must be at least 8 characters"

Definition of Done:

- ☒ Flutter UI complete (signup form, validation)
 - ☒ Backend API: POST /api/v1/auth/signup implemented
 - ☒ Database: User table with email, password_hash, created_at
 - ☒ Password hashing: bcrypt with cost factor 12
 - ☒ Unit tests: 5 test cases (valid, duplicate, weak password, etc.)
 - ☒ Integration tests: End-to-end signup flow
 - ☒ Code reviewed and approved
 - ☒ Merged to main branch
-

US-002: User Login

Title: As a registered user, I want to log in so that I can access my account

Priority: CRITICAL

Story Points: 3

Acceptance Criteria:

- Given: User has valid account
- When: Enters email and password, clicks Login
- Then: User logged in, JWT token created, home screen displayed
- Given: User enters invalid email/password
- When: Clicks Login
- Then: Error "Invalid email or password" (don't reveal which)

- Given: User enters 5 incorrect passwords in 15 minutes
- When: Attempts 6th login
- Then: Account temporarily locked, message "Too many attempts, try again in 15 minutes"

Definition of Done:

- ☒ Flutter UI: Login form
 - ☒ Backend API: POST /api/v1/auth/login
 - ☒ JWT generation and signing
 - ☒ Rate limiting: 5 failures per 15 minutes
 - ☒ Unit tests: 4 test cases
 - ☒ Code reviewed
 - ☒ Merged to main
-

US-003: Initiate Call

Title: As a user, I want to call a contact so that I can have a video/audio conversation

Priority: CRITICAL

Story Points: 13

Acceptance Criteria:

- Given: User logged in, has contacts
- When: Clicks on contact name, then "Call" button (video or audio)
- Then: Outgoing call UI displayed, ringing, call state tracked
- Given: Call initiated
- When: Recipient online
- Then: Recipient receives notification (sound + visual), can accept or decline
- Given: Call initiated
- When: Recipient offline

- Then: Call invitation queued, will retry when recipient comes online
- Given: Recipient accepts call
- When: WebRTC connection established
- Then: Both participants see video, hear audio, call duration timer starts

Definition of Done:

- ☒ Flutter UI: Call screen with video preview, controls
 - ☒ WebRTC: flutter_webrtc integration, ICE candidates gathered
 - ☒ Backend: WebSocket signaling for call invite/accept/decline
 - ☒ Mediasoup: Producer/Consumer setup for media
 - ☒ SRTP: Encryption verified
 - ☒ Recording: Started automatically
 - ☒ Unit tests: 8+ test cases
 - ☒ Integration tests: End-to-end call flow
 - ☒ Performance: <5 second call establishment measured
 - ☒ Code reviewed
 - ☒ Merged to main
-

US-004: End Call

Title: As a user in a call, I want to end the call so that I can disconnect

Priority: HIGH

Story Points: 3

Acceptance Criteria:

- Given: User in active call
- When: Clicks "End Call" button
- Then: Call terminated, both participants disconnected, recording saved
- Given: Call ended
- When: User reviews call history

- Then: Call listed with correct duration

Definition of Done:

- ☒ Flutter UI: End call button accessible during call
 - ☒ WebRTC: Close connections cleanly
 - ☒ Mediasoup: Close producer/consumer
 - ☒ Recording: Finalized and saved to disk
 - ☒ Backend: Call state updated, end timestamp recorded
 - ☒ Unit tests: 3 test cases
 - ☒ Code reviewed
 - ☒ Merged to main
-

US-005: View Admin Dashboard

Title: As an admin, I want to view the admin dashboard so that I can manage users and calls

Priority: CRITICAL

Story Points: 13

Acceptance Criteria:

- Given: Admin user logged in
- When: Clicks Settings → Admin Dashboard (admin-only visible)
- Then: Dashboard loaded with tabs: Users, Calls, Recordings, Analytics
- Given: On Users tab
- When: Dashboard loads
- Then: User list displayed (email, name, status, registration date, last login)
- Given: User list displayed
- When: Admin clicks "Create User"
- Then: Dialog opened to enter email, auto-generate temporary password

- Given: User list displayed
- When: Admin clicks "Suspend" on a user
- Then: User status changed to "suspended", user cannot call
- Given: On Calls tab
- When: Dashboard loads
- Then: Call list displayed (date, participants, duration, status) with metrics (total calls, success rate, avg duration)
- Given: On Recordings tab
- When: Dashboard loads
- Then: Recording list displayed (date, participants, size) with download/delete options
- Given: On Analytics tab
- When: Dashboard loads
- Then: Graphs displayed (latency, jitter, packet loss, MOS scores) with alerts for poor calls

Definition of Done:

- ☒ Flutter UI: Admin dashboard with 4 tabs
 - ☒ Backend API: GET /api/v1/admin/users, /calls, /recordings, /metrics
 - ☒ Admin-only access: Verify role before serving data
 - ☒ Database queries: Optimized for fast load (<2 seconds)
 - ☒ Audit logging: All admin actions logged
 - ☒ Unit tests: 20+ test cases for CRUD operations
 - ☒ Integration tests: End-to-end admin workflows
 - ☒ Code reviewed
 - ☒ Merged to main
-

5. ACCEPTANCE CRITERIA SUMMARY

Must Pass Before Launch (September 10):

Functional Testing:

- ☒ Signup: Create account, valid email/password
- ☒ Login: Login with email/password, JWT token issued
- ☒ Logout: Session invalidated
- ☒ Call: Initiate, receive, accept, decline, end calls
- ☒ Recording: Automatic MP4 creation, 1-month retention
- ☒ Contacts: Add, view, call from contact
- ☒ Settings: Change device settings, quality settings
- ☒ Admin: User CRUD, call history view, recording download, analytics

Non-Functional Testing:

- ☒ Security: SRTP encryption verified, passwords hashed, tokens secure
- ☒ Performance: <5s call setup, <300ms latency (95th %ile), >4.0 MOS score
- ☒ Reliability: <2% failure rate on calls, graceful error handling
- ☒ Usability: Simple UI, no training required, <100ms button response
- ☒ Cameroon-specific: Adaptive bitrate, FEC, offline queue, packet loss resilience

Network Testing (Critical):

- ☒ Good network (10 Mbps, <50ms RTT, <0.1% loss): Full HD, no issues
- ☒ Fair network (2.5 Mbps, 80ms RTT, <1% loss): 720p, smooth audio
- ☒ Poor network (0.5 Mbps, 150ms RTT, 5% loss): 240p or audio-only, usable
- ☒ Packet loss (3-5% simulated): Call quality maintained
- ☒ High latency (200ms+): Audio quality acceptable despite delay

Test Coverage:

- ☒ Code coverage: >80% (unit tests)
 - ☒ Integration tests: Call flows, authentication, recording
 - ☒ System tests: Performance, stress (20 concurrent users), Cameroon scenarios
 - ☒ User acceptance: Internal team calls successfully
-

6. PHASE BREAKDOWN (2 Months, 4 Sprints)

Sprint 1 (Sep 1-5): Foundation

- Authentication (signup, login, logout)
- User database schema
- Admin dashboard scaffolding
- WebSocket signaling setup

Sprint 2 (Sep 6-10): MVP Features

- One-on-one calls (audio + video)
- Call recording
- Contacts management
- Settings (device config)
- Admin: User management, call history

Sprint 3 (Sep 11-25): Polish & Testing

- Performance optimization (latency, bitrate adaptation)
- Bug fixes & stabilization
- Cameroon network testing (3-5% loss, poor latency scenarios)
- User acceptance testing with internal team
- Security audit (SRTP, password hashing)

Sprint 4 (Sep 26-10): Launch & Monitoring

- Deployment to production
 - Admin monitoring dashboard
 - Analytics collection (MOS, latency, jitter)
 - Support documentation
 - Launch Day: October 10 (Sep 10 in original planning might mean mid-September)
-

7. SUCCESS METRICS (Launch Day & Beyond)

Immediate (Launch):

-  20 concurrent users supported
-  Zero critical bugs
-  >99% call success rate
-  <5s call setup time
-  MOS > 4.0 on good/fair networks

Month 1-2:

-  >100 registered users
-  <2% call failure rate
-  99.5% platform uptime
-  <1% security incidents

Year 1:

-  400,000 registered users
-  1,000+ concurrent calls capacity
-  <1% churn rate
-  >4.5 stars user rating

8. DEFINITIONS & ABBREVIATIONS

Term	Definition
E2E	End-to-End encryption
FEC	Forward Error Correction (redundancy for packet loss)
JWT	JSON Web Token (authentication token)
MOS	Mean Opinion Score (call quality 1-5)
MVP	Minimum Viable Product (launch features)

Term	Definition
P2P	Peer-to-Peer (direct connection)
QoS	Quality of Service (network prioritization)
RFC	Request for Comments (IETF standards)
RTP	Real-time Transport Protocol (media transport)
RTT	Round-Trip Time (latency both directions)
SRTP	Secure RTP (encrypted media)
SFU	Selective Forwarding Unit (media relay)
WebRTC	Web Real-Time Communication (P2P media)
WSS	WebSocket Secure (encrypted WebSocket)

9. DOCUMENT APPROVAL

Role	Name	Signature	Date
Product Manager	[TBD]	_____ —	_____
CTO	[TBD]	_____ —	_____
QA Lead	[TBD]	_____ —	_____
Project Manager	[TBD]	_____ —	_____

END OF IEEE 830 SRS DOCUMENT

Version: 1.0

Last Updated: August 24, 2026

Status: READY FOR DEVELOPMENT

This SRS is now **APPROVED FOR DEVELOPMENT**. All teams should reference this document as the single source of truth for requirements.